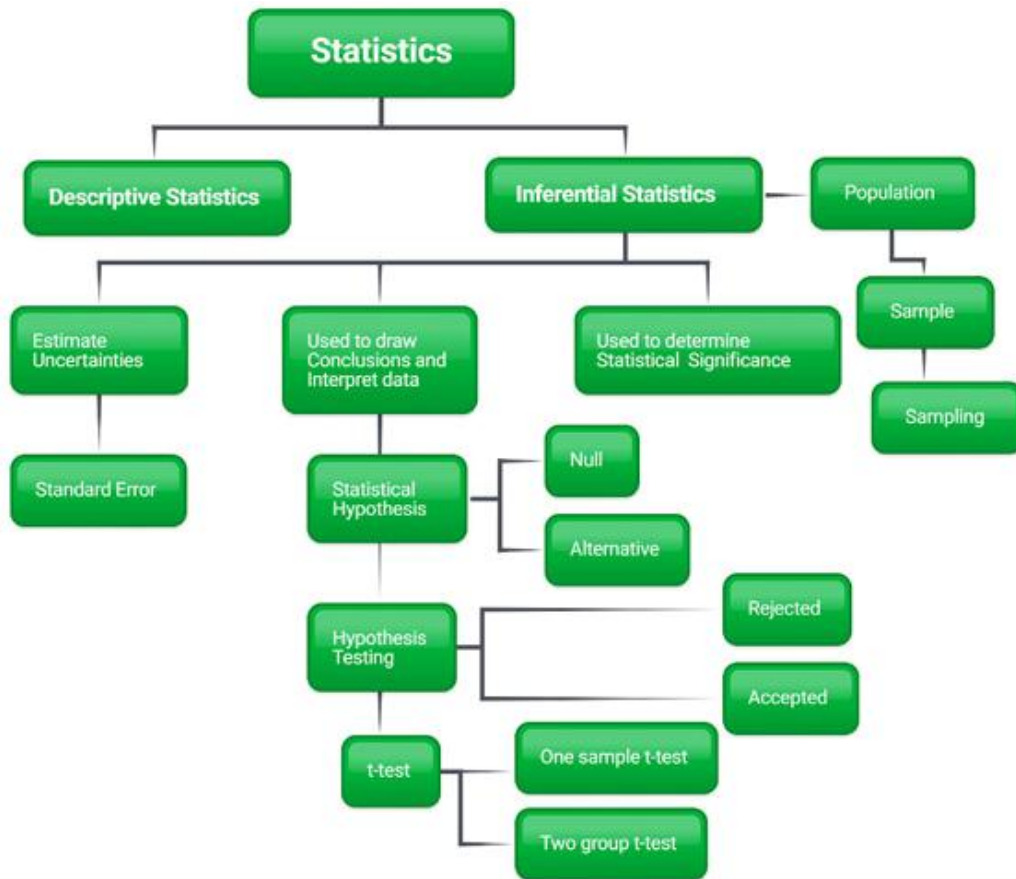


SAMPLING AND ESTIMATION



INFERENCEAL STATISTICS

Inferential statistics allows us to make predictions from the data.

Inferential Statistics involves using sample data to make generalizations or predictions about a larger population. Instead of analyzing an entire population (which is often impractical or impossible), inferential statistics uses a sample to infer conclusions and assess how accurate these conclusions are.

Components of Inferential Statistics

- **Population:** The entire group you want to study (e.g., all customers of a store).
- **Sample:** A smaller, representative group taken from the population (e.g., 100 customers surveyed).
- **Estimation:** Estimate population parameters (e.g., mean, proportion) using sample statistics.
- **Confidence Intervals:** A range of values likely to contain the population parameter (e.g., "The average height is between 160 cm and 170 cm with 95% confidence").

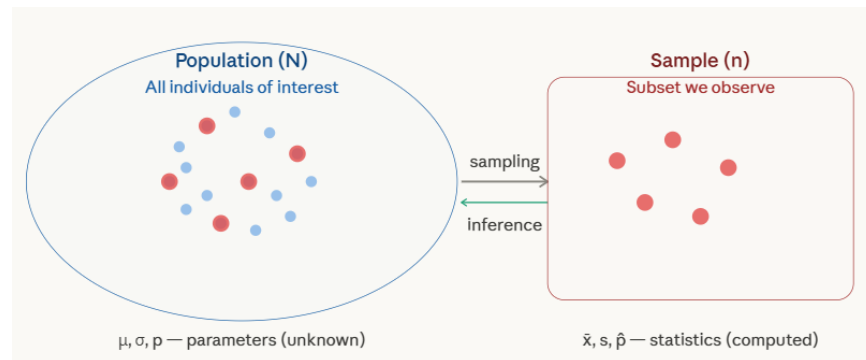
- **Hypothesis Testing:** Testing a claim about the population using sample data (e.g., t-tests, chi-square tests).
- **Regression Analysis:** Predicting relationships between variables (e.g., sales and marketing spend).
- **ANOVA (Analysis of Variance):** Comparing means across multiple groups.

POPULATION AND SAMPLE

Population refers to group of objects, events, experiments, observations, procedures etc.

- Finite Population:** It is a group of objects, events, observations which are countable
Ex: no. of workers in a factory.
- Infinite Population :** It is a group of objects, events, observations which are uncountable
Ex: All stars in the milky way galaxy.

Sample is the smaller group of members of the selected population to represent the population.



	POPULATION	SAMPLE
1.	Population is a collection of all elements having same characteristics.	Sample is a subgroup of members of the population i.e. a sample consists of one or more observations drawn from the population.
2.	The measurable characteristic of population is called parameter.	The measurable characteristic of sample is called statistic.
3.	A survey done of the entire population is accurate and more precise with no margin of error.	A survey done using a sample of the population has a margin of error.
4.	The population focuses on the identification of the characteristic.	The sample makes inference about the population

SAMPLING

There are two types of sampling.

1. Probability Sampling / Non- Probability Sampling
2. Univariate Sampling / Bivariate Sampling

PROBABILITY SAMPLING (UNBIASED SAMPLING)

In this sampling the selection of members from the population is random such that **each member has an equal chance of being selected**.

There are 4 types of Probability Sampling:

1. Simple Random Sampling

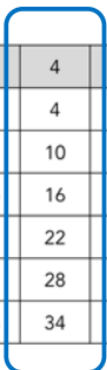
It is the most common method of sampling where every member of the population has equal chance of being selected.

Ex: Selecting 100 employees from 1000 employees in a factory.

2. Systematic Sampling:

Every member is assigned a number but instead of randomly selecting numbers, the numbers are chosen at regular intervals.

Ex: Out of 1000 employees, starting from member with number 4, every 10th member is selected. So members with numbers 4, 14, 24.... will be selected.

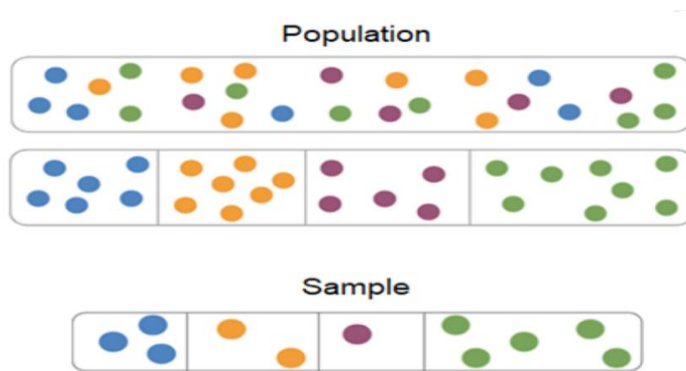


1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36

3. Stratified Sampling:

Population is divided into subgroups called strata, based on relevant characteristics like age, gender, income etc. The members from each subgroup are selected using simple random or systematic sampling.

Ex: From 700 males and 300 females, 100 employees to be selected reflecting gender balance. Divide employees in subgroups of only males and only females. Select 70 males and 30 females using simple random or systematic sampling.



In **stratified sampling**, you divide the population into subgroups (strata) and take samples from all strata.

4. **Cluster Sampling:**

Population is divided into subgroups with similar characteristics of the whole population

Ex: Imagine Sunway Medical Centre wants to survey patients about their satisfaction levels. Here's how they could use cluster sampling:

1. **Population:** All patients treated at the hospital in the last month.
2. **Clusters:** Group patients by the ward they were treated in (e.g., Ward A, Ward B, Ward C).
3. **Randomly Select Clusters:** Randomly pick two wards (e.g., Ward A and Ward C).
4. **Survey Everyone in selected Clusters:** Survey all patients in Ward A and Ward C.

In **cluster sampling**, you divide the population into clusters and take entire clusters as samples.

NON- PROBABILITY SAMPLING (BIASED SAMPLING)

In this type of sampling the selection of members from population is non-random and each member does not have equal chance of being selected.

There are 4 types:

1. **Convenience Sampling:**

Members are selected which are most convenient for researcher.

Ex: while researching about the services provided by the university to the hostel students, you can ask your fellow student to complete the survey.

2. **Voluntary response sampling:**

It is similar to convenience sampling, but here people who are themselves ready to conduct the survey collects the sample data.

3. **Judgement Sampling:**

This sampling is based on the opinion of an expert. The quality of the sample results depends on the judgement of the person.

Ex: selection procedure of all the reality shows.

4. Snowball Sampling:

In this method, we first select some persons, then with the help of selected persons we select some more and this process continues to collect the sample. It is same as snowball increases in size as it rolls down.

Ex: while researching about child labours, you meet one child who connects you with other child labours.

Univariate Sampling:

In univariate sampling, data is collected for **a single characteristic** or variable.

- A **univariate sample**: you might take a random sample of 100 people and record only their **age**. That dataset is univariate.
- Univariate data helps to analyse mean, median, mode, variance, standard deviation.
- Univariate data can be represented as histogram, bar chart, pie chart, box plot

Bivariate Sampling:

In bivariate sampling, data is collected for **two related variables from each sample unit**, to study relationship between them. It is used to study concepts like **correlation and regression**.

A **bivariate sample**: you sample 100 students and for each one you record study hours and exam score. Now each person gives you a pair (X, Y) . That's bivariate data.

Student	Study Hours	Exam Score
A	2	60
B	4	75
C	6	90

Bivariate data helps to examine **association**, find **correlation** and dependence between variables to make inference about **relationship in the population**.

Bivariate data can be represented as Scatter plot, Correlation analysis, Regression lines

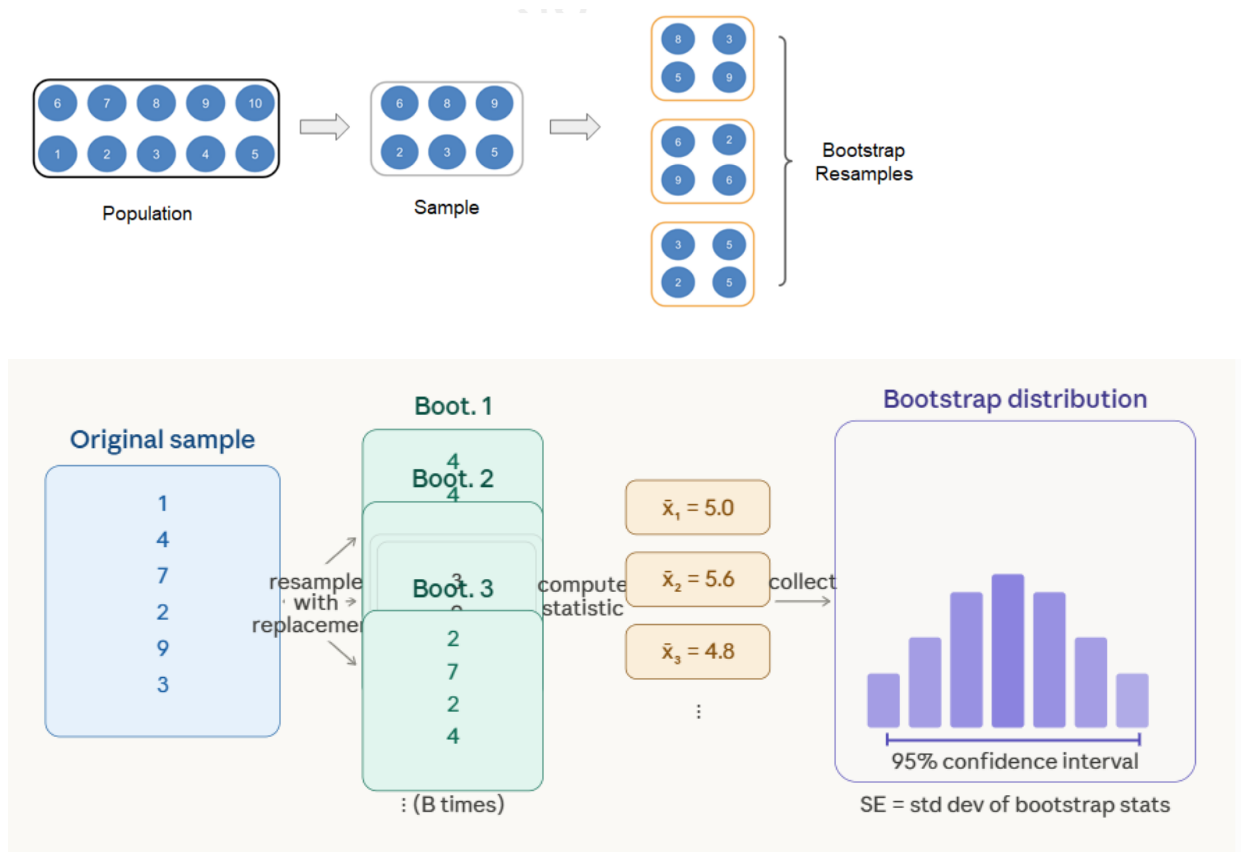
RESAMPLING

Resampling is a technique in which we draw a sample from the available sample.

The Resampling techniques are:

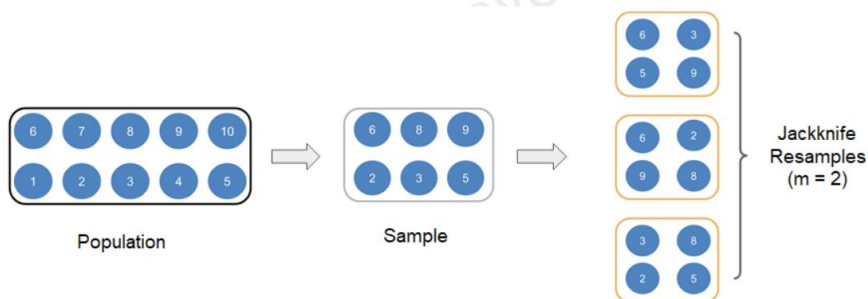
1. **Bootstrap Resampling:**

It is a simple random sample with replacement, drawn from a sample. Bootstrap lets you estimate the standard error and confidence intervals for any statistics.



2. Jackknife Resampling:

It is a method in which one deletes m observations from the original sample.



PARAMETER AND STATISTIC

Parameter:

A measurable characteristics of a population is called parameter.

Eg. Population mean(μ), population variance(σ^2), population standard deviation (σ)

A parameter is used to describe the entire population.

Statistic:

A measurable characteristics of a sample is called a statistics.

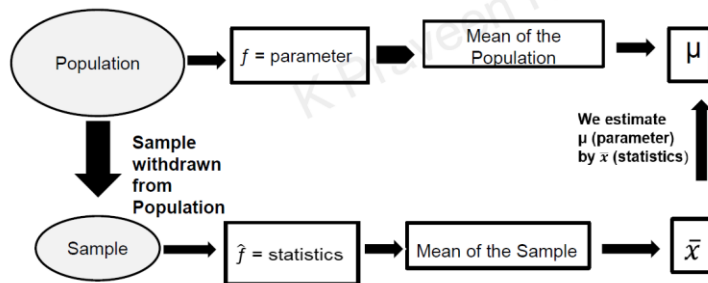
Eg: Sample mean ($\bar{x}$), sample variance (s^2), sample standard deviation(s)

In Inferential statistics we use sample statistics to estimate the corresponding population parameter.

Ex: We catch 100 house sparrows and measure their length. The mean length of 100 sparrows is a statistic which can be used to make an inference about the length of the entire house sparrow population.

Relation between Parameter and Statistics

- Considering mean as the population Parameter



SAMPLING DISTRIBUTION

A **sampling distribution** is a statistical concept that describes the distribution of a specific sample statistic (like mean, proportion, or variance) across multiple samples drawn from the same population. It represents the variability of the statistic across different samples.

If you repeatedly take random samples from the population and calculate a statistic for each sample, the resulting set of statistics forms the sampling distribution.

- The sampling distribution of the **sample mean** shows how the sample mean varies from sample to sample.
- The sampling distribution of the **sample proportion** shows how the sample proportion varies across samples.

A sample proportion is the proportion (or fraction) of a certain characteristic observed in a sample. i.e. x/n (required/total)

The Shape of Sampling Distribution depends on Sample Size:

- For large enough sample sizes, the sampling distribution of the mean tends to be approximately normal (Central Limit Theorem).
- Smaller sample sizes may result in less predictable shapes.

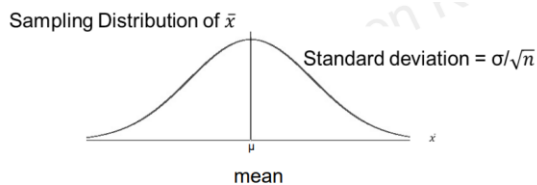
Characteristics of Sampling Distribution

1. **Mean:** The mean of the sampling distribution equals the population mean (μ)
2. **Standard Deviation :** The standard deviation of the sampling distribution (called the standard error) depends on the population standard deviation (σ) and the sample size (n).

$$\text{Standard Error} = \frac{\sigma}{\sqrt{n}}$$

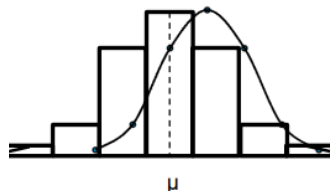
Larger sample sizes lead to smaller standard errors, making the sampling distribution narrower.

3. **Shape:** For large sample sizes, the sampling distribution of the mean becomes approximately normal, regardless of the population's distribution (Central Limit Theorem).



Imagine Sunway Medical Centre wants to study the average recovery time of patients after a specific surgery.

1. **Population:** All patients who underwent the surgery.
 - True average recovery time (population mean) = **10 days**.
2. **Sampling:** Take multiple random samples of **50 patients** each.
 - For each sample, calculate the **sample mean recovery time**.
3. **Sampling Distribution:**
 - Plot all the sample means (one from each sample) to create the sampling distribution of the sample mean.
 - The shape of this distribution will indicate how sample means are distributed around the population mean.



Example of sampling distribution

1. Sample Mean ($\bar{x}$)

- **Population:** The heights of all adults in a city (mean height $\mu=170$ cm, standard deviation $\sigma=10$ cm).
- **Process:** Take multiple random samples of size 30 and calculate the mean height for each sample.
- **Result:** The sample means ($\bar{x}_1, \bar{x}_2, \dots$) will form a sampling distribution.
 - Mean of the sampling distribution = Population mean ($\mu=170$).
 - Standard deviation of the sampling distribution (Standard Error) = $\frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{30}}$

$$= 1.83$$

2. Sample Proportion (p)

- **Population:** 60% of voters support a candidate ($p=0.6$).
- **Process:** Take random samples of 100 voters and calculate the proportion of supporters in each sample.
- **Result:** The sample proportions ($p_1, p_2, \dots$) will form a sampling distribution.
 - Mean of the sampling distribution = Population proportion ($p=0.6$).
 - Standard deviation (Standard Error) = $\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.6(1-0.6)}{100}} = 0.049$

3. Sample Variance (s^2)

- **Population:** The weights of apples in a basket ($\sigma^2 = 4\text{kg}$).
- **Process:** Take random samples of 20 apples and compute the variance of weights for each sample.
- **Result:** The sample variances ($s_1^2, s_2^2, \dots$) will form a sampling distribution.
 - The mean of the sampling distribution of s^2 will approach $\sigma^2=4$

CENTRAL LIMIT THEOREM

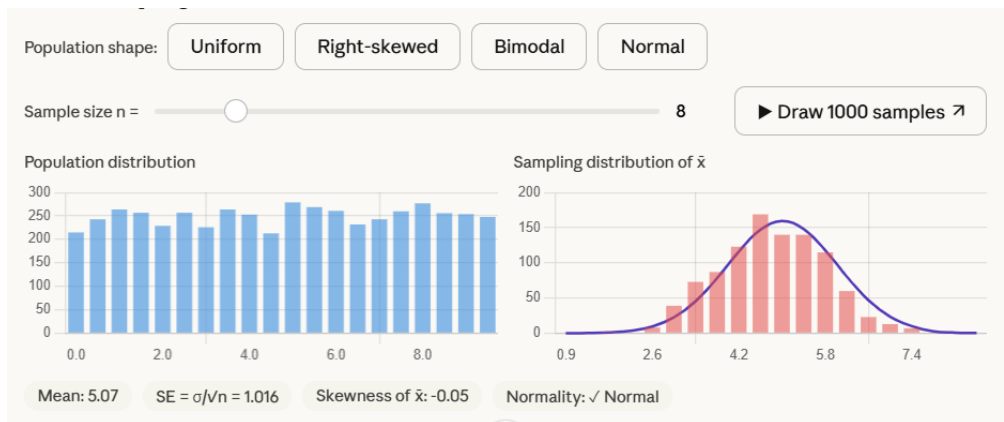
Central Limit Theorem (CLT) states that regardless of the population's shape, the sampling distribution of the sample mean **approaches a normal distribution** (bell shaped curve) as sample size grows. A sample size of 30 or more is considered to be sufficient to hold CLT and as the sample size becomes large the prediction of characteristics of population becomes more accurate. CLT is powerful as even if your data is Skewed, Non-normal, or noisy, sample means will still form a Normal Distribution.

Ex: A coffee shop daily sales are highly irregular. If we take 100 samples and compute mean each time then distribution of those means becomes **normal**.

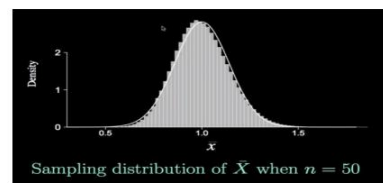
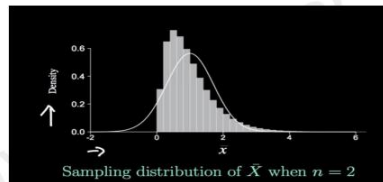
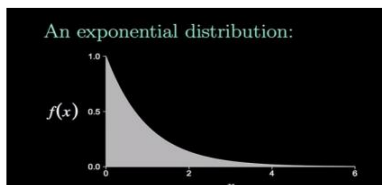
Magic of CLT: Visual of Original data is skewed, but of sample means is a normal bell shaped curve.



central_limit_theorem_interactive.html



Central Limit Theorem – Graph based on samples



Sample mean to be appr. Normal distributed, sample size should be ≥ 30

STATISTICAL INFERENCE

Statistical Inference is a process through which inferences about the population are made based on information obtained from the sample.

(When you are ready to cross a street, you estimate the speed of any car that is approaching. You know that the speed you estimated is not exact but allow for a margin of error to be safe while crossing)

There are mainly 3 types of statistical inferences.

1. Point Estimation
2. Interval Estimation
3. Hypothesis Testing

POINT ESTIMATION

Point Estimation involves the use of sample data to calculate a single value of an unknown population parameter. Single value of sample data is known as Point Estimate of corresponding population parameter.

For example, sample mean $\bar{x}$ is a point estimate of population mean μ , sample proportion $\bar{p}$ is a point estimate of population proportion p .

Suppose the population consists of N objects and the sample consists of n objects. Also all possible samples of n objects are equally likely to occur, then

$$\text{Sample mean} = \frac{\sum x}{n} \qquad \text{Sample proportion } \bar{p} = \frac{x}{n}$$

$$\text{Sample standard deviation 's'} = \frac{\sqrt{\sum (x - \mu)^2}}{n - 1}$$

The error in estimation, using $\bar{x}$ to estimate μ is given by standard deviation of sampling distribution called standard error. If this standard deviation is large we are less precise in estimating μ .

$$\text{Standard Error} = \frac{\sigma}{\sqrt{n}}$$

Example of Point Estimation

Imagine Sunway Medical Centre wants to estimate the **average recovery time** for patients who underwent a specific surgery.

1. **Population Parameter:** The true average recovery time (μ) for all patients who had the surgery.
2. **Sample Data:** A random sample of 50 patients is selected, and their average recovery time is found to be **8 days**.
3. **Point Estimate:** The sample mean ($\bar{x}=8$) is used as the point estimate for the population mean (μ).

Interpretation: Based on the sample data, the hospital estimates that the average recovery time for all patients is approximately **8 days**.

Example:

In a survey question for a sample of 200 individuals, 80 persons gave response 'Yes', 100 gave response 'No' and 20 gave no response.

a) What is the point estimate of the proportion in the population who respond 'Yes'?

b) What is the point estimate of the proportion in the population who respond 'No'?

Given $n = 200$, $X_{\text{yes}} = 80$, $X_{\text{no}} = 100$, $X_{\text{none}} = 20$

1. **point estimate** of the proportion in the population who respond 'Yes'

$$\hat{p} = \frac{x}{n} = \frac{80}{200} = 0.4$$

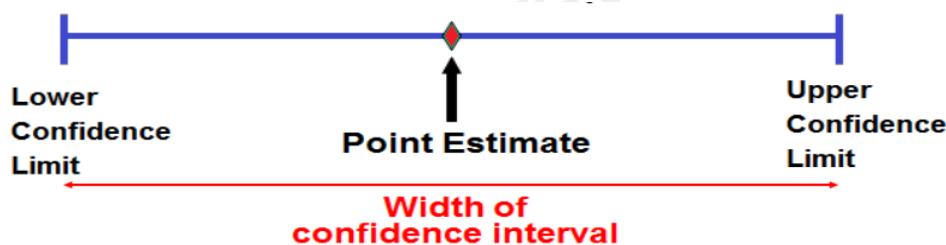
2. **Point estimate** of the proportion in the population who respond 'No' = $\frac{100}{200} = 0.5$

INTERVAL ESTIMATION

In Interval Estimation we find two numbers using sample statistic between which a population parameter lie. For example, $a < \mu < b$ is an interval estimate of the population mean μ . It indicates that population mean is greater than a but less than b .

Confidence Interval:

A **Confidence Interval (CI)** is a range of values used to estimate a population parameter (like the mean or proportion) with a certain level of confidence. Instead of giving a single value (point estimate), it provides a range within which the parameter is likely to fall.

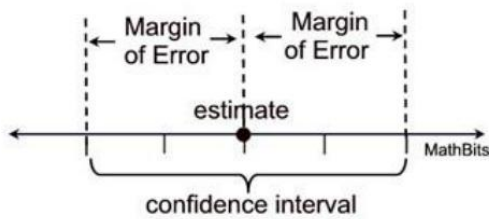


Features of Confidence Intervals

1. **Interval Estimate:** It is a range of values of sample statistics used to estimate an unknown population parameter.

In interval estimate there is upper limit, lower limit

2. **Confidence Level:** The **probability part** of a confidence interval is called **confidence level**. It indicates how confident we are that the interval contains the true population parameter. Common confidence levels are 90%, 95%, and 99%.
 - A 95% confidence level means that if we repeated the sampling process many times, about 95% of the calculated intervals would contain the true parameter.
 - Confidence level written in decimal is called **confidence coefficient**.
3. **Margin of Error:** In a confidence interval, the range of values, above and below the sample statistic is called margin of error. It reflects the uncertainty in the estimate and is influenced by the sample size, variability, and confidence level.



Point Estimate	Confidence Level	Margin of Error
$\mu = \bar{x}$	$Z_{\frac{\alpha}{2}}$	$\pm \frac{\sigma}{\sqrt{n}}$

Ex: Suppose a news channel conducted an exit poll and reports that candidate X will receive 40% of the vote. The channel claimed that exit poll had 5% margin of error and confidence level is 90%. It means channel is 90% confident that candidate X will get between 35% and 45% of the vote.

Interval Estimation of Population Mean

An **interval estimate** gives a range within which the true population mean μ is expected to lie.

In order to develop an interval estimate of a population mean, the **population standard deviation σ** or **sample standard deviation 's'** must be known to compute the margin of error.

population mean = point estimate $\pm$ margin of error

$$\mu = \bar{x} \pm \text{margin of error}$$

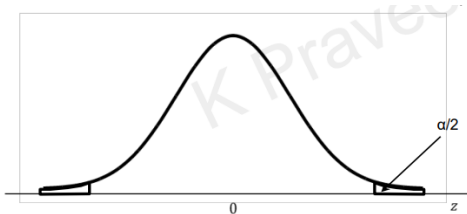
$$p = \bar{p} \pm \text{margin of error}$$

The **margin of error** tells how much uncertainty is allowed in the estimate.

If population standard deviation σ is known OR If sample standard deviation s is known

$$\text{Margin of error} = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \quad \text{OR} \quad \text{Margin of error} = Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

where $(1 - \alpha)$ is called **confidence coefficient** and $Z_{\alpha/2}$ is **critical Z value** providing an area of $\frac{\alpha}{2}$ in the upper tail of the standard normal probability distribution.



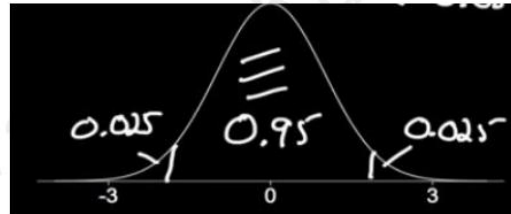
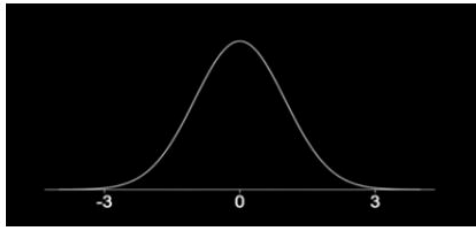
For a 95% confidence interval:

- $\alpha = 0.05$
- $\alpha/2 = 0.025$
- $Z_{0.025} = 1.96$

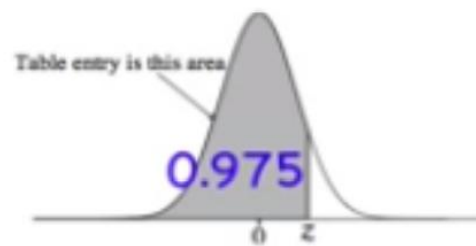
The values of $Z_{\alpha/2}$ for most commonly used levels is given below:

Confidence level	A	$\alpha/2$	$Z_{\alpha/2}$
90%	0.10	0.05	1.645
95%	0.05	0.025	1.960
99%	0.01	0.005	2.576

What is the approximate z value for a 95% confidence Interval ?



$(1 - \alpha) 100\% = 95\%$
 $100\% - 100\alpha\% = 95\%$
 $-100\alpha\% = -5\%$
 $\alpha = 5/100 = .05$
 $\alpha = 0.05$
 Area to the left of the required z value = $1 - 0.025$
 $= 0.975$
 $Z_{\alpha/2}$ (critical value) = 1.96



Example:

Suppose Sunway Medical Centre wants to estimate the **average recovery time** for patients after a specific surgery:

1. Sample Data:

- Sample size (n) = 50 patients
- Sample mean ($\bar{x}$) = 8 days
- Sample standard deviation (s) = 2 days
- Confidence level = 95% (corresponds to $z=1.96$)

2. Calculate Margin of Error:

$$\text{Margin of Error} = z \cdot \frac{s}{\sqrt{n}} = 1.96 \cdot \frac{2}{\sqrt{50}} = 0.55$$

3. Confidence Interval:

$$CI = \bar{x} \pm \text{Margin of Error} = 8 \pm 0.55$$

$$CI = (7.45, 8.55)$$

4. **Interpretation:** With 95% confidence, the **true average recovery time** for all patients lies between **7.45 and 8.55 days**.

Example:

Suppose a student measuring the boiling temperature of a certain liquid observes the reading (in degree Celsius) 102.5, 101.7, 103.1, 100.9, 100.5, 102.2 on 6 different samples of the liquid. If he knows that the standard deviation of this procedure is 1.2°C, what is the interval estimation for the population mean at a 95% confidence level?

$$n = 6, \sigma = 1.2, \text{ confidence level} = 95\%$$

$$\text{mean} = (102.5 + 101.7 + 103.1 + 100.9 + 100.5 + 102.2) / 6 = 101.82$$

$$\text{confidence level} = 95\%$$

$$(1 - \alpha) = 0.95, \alpha = 0.05, \alpha/2 = 0.025$$

$$Z_{0.025} = 1.96$$

$$\text{Margin of error} = Z_{0.025} \cdot \frac{1.2}{\sqrt{6}} = 0.96$$

$$\mu = \bar{x} \pm \text{margin of error} = 101.82 \pm 0.96$$

$$\text{so confidence interval is } (101.82 - 0.96, 101.82 + 0.96) = (100.86, 102.78)$$

INTERVAL ESTIMATION OF POPULATION PROPORTION

The interval estimation of population proportion is given by :

$$\bar{p} \pm \text{margin of error}, \text{ where margin of error} = Z_{\alpha/2} \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

Example:

A simple random sample of 400 individuals provides 100 Yes responses.

- 1 .What is the point estimate of the population proportion that would provide Yes response?
- 2 .Compute the 95%confidence interval for the population proportion.

Solution:

$$n = 400, X_{\text{yes}} = 100$$

- 1.Point estimate of the proportion in population who respond Yes

$$\bar{p} = \frac{X_{\text{yes}}}{n} = \frac{100}{400} = 0.25$$

- 2.Given $(1 - \alpha) = 0.95 \Rightarrow \alpha/2 = 0.025$

$$Z_{0.025} = 1.96, \text{ margin of error} = 1.96 \cdot \sqrt{\frac{0.25(1-0.25)}{400}} = 0.0424$$

$$p = 0.25 \pm 0.0424$$

confidence interval is $(0.25 - 0.0424, 0.25 + 0.0424) = (0.2076, 0.2924)$

When we have a sample of size < 30 , population follows normal distribution and standard deviation is unknown, in order to estimate population mean we use t- distribution with $n-1$ degrees of freedom.

Interval Estimation Procedure

